diatomic gas molecules by 1°C it require more heat energy than monoatomic molecules.

$$\therefore \gamma = \frac{C_P}{C_V} = \frac{\frac{9}{2}R}{\frac{7}{2}R} = \frac{9}{7} = 1.28$$

iii) Triatomic molecule

a) Linear molecule

Energy of one mole= $\frac{7}{2}kT \times N_A = \frac{7}{2}RT$

$$C_V = \frac{dU}{dT} = \frac{d}{dT}\left[\frac{7}{2}RT\right]$$

$$C_V = \frac{7}{2}R$$

$$C_P = C_V + R = \frac{7}{2}R + R = \frac{9R}{2}$$

$$\therefore \gamma = \frac{C_P}{C_V} = \frac{\frac{9}{2}R}{\frac{7}{2}R} = \frac{9}{7} = 1.28$$

b) Non-linear molecule

Energy of a mole = $\frac{6}{2}kT \times N_A = \frac{6}{2}RT = 3RT$

$$C_V = \frac{dU}{dT} = 3R$$

$$C_P = C_V + R = 3R + R = 4R$$

$$\therefore \gamma = \frac{C_P}{C_V} = \frac{4R}{3R} = \frac{4}{3} = 1.33$$

Note that according to kinetic theory model of gases the specific heat capacity at constant volume and constant pressure are independent of temperature. But in reality it is not sure. The specific heat capacity varies with the temperature.

EXAMPLE 9.5

Find the adiabatic exponent γ for mixture of μ_1 moles of monoatomic gas and μ_2 moles of a diatomic gas at normal temperature (27°C) .

Solution

The specific heat of one mole of a monoatomic gas $C_V = \frac{3}{2}R$

For μ_1 mole, $C_V = \frac{3}{2}\mu_1 R \quad C_P = \frac{5}{2}\mu_1 R$

The specific heat of one mole of a diatomic gas

$$C_V = \frac{5}{2}R$$

For μ_2 mole, $C_V = \frac{5}{2}\mu_2 R \quad C_P = \frac{7}{2}\mu_2 R$

The specific heat of the mixture at constant volume $C_V = \frac{3}{2}\mu_1 R + \frac{5}{2}\mu_2 R$

The specific heat of the mixture at constant pressure $C_P = \frac{5}{2}\mu_1 R + \frac{7}{2}\mu_2 R$

The adiabatic exponent $\gamma = \frac{C_P}{C_V} = \frac{5\mu_1 + 7\mu_2}{3\mu_1 + 5\mu_2}$

9.5 MEAN FREE PATH

Usually the average speed of gas molecules is several hundred meters per second even at room temperature (27°C). Odour from an open perfume bottle takes some time to reach us even if we are closer to the room. The time delay is because the odour of the molecules cannot travel straight to us as it undergoes a lot of collisions with the nearby air molecules and moves in a zigzag path. This *average distance travelled by the molecule between two successive collisions is called mean free path (λ). We can calculate the mean free path based on kinetic theory.*

Expression for mean free path

We know from postulates of kinetic theory that the molecules of a gas are in random motion and they collide with each other. Between two successive collisions, a molecule

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